

Jiri Musil

Coursework, labs, and projects completed in *Spring 2026*. Neural networks, transformers, diffusion, reinforcement learning, and a final multi agent system.

→ [GITHUB REPOSITORY](#)

[github.com/JiriCZTX/Jiri_Musil_ITAI_2376_Deep_Learning](#)

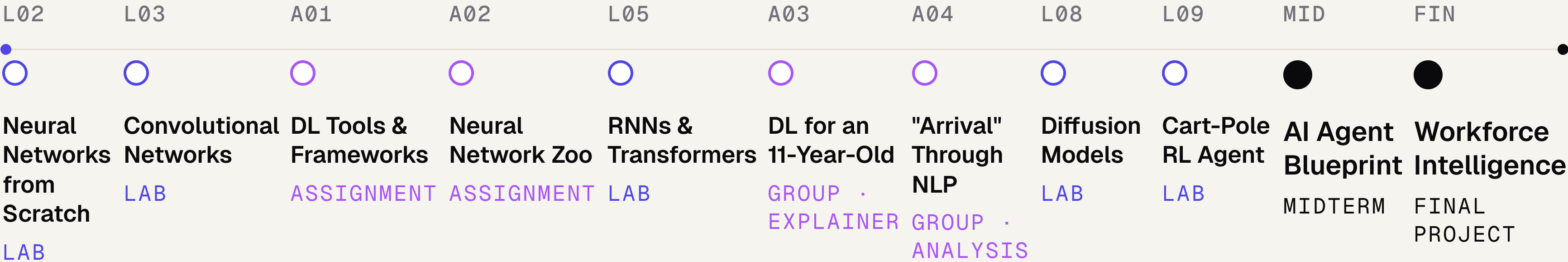
Public · [README](#) · [Code](#) · [Notebooks](#) · [Slides](#)

PROGRAM	Applied AI & Robotics, <i>HCC</i>
FOCUS	Multi-agent systems · NLP · forecasting
STACK	PyTorch · HF Transformers · CrewAI
STATUS	Open to roles & collaboration

• THE COURSEWORK

Eleven modules across the *deep learning* curriculum, with two capstone projects.

Spring 2026 · 16 weeks
Houston Community College
Instructor: Patricia McManus



○ Lab notebook ○ Assignment ● Capstone

6+
HANDS-ON LABS

4
WRITTEN DELIVERABLES

2
CAPSTONE PROJECTS

Workforce Intelligence.

A *two agent* system for talent and attrition.

AGENT 01 • TALENT INTELLIGENCE

Routed NER ensemble + SBERT semantic match.

DistilBERT • GLiNER • ModernBERT
10 entity types • curated gazetteer

PROPRIETARY BRAIN



6 scenarios
CrewAI + Claude (joint hire)

AGENT 02 • WORKFORCE FORECASTING

Bi-LSTM attrition & headcount projection.

Bi-LSTM attrition model
Temporal patterns • risk alerts

A dashboard that *routes* entity recognition across DistilBERT, GLiNER, and ModernBERT, matches candidates to roles with SBERT, and forecasts department attrition with a Bi-LSTM.

- 01 Routed NER ensemble • model selection per entity class
- 02 SBERT job to candidate matching • cosine similarity over embeddings
- 03 Bi-LSTM attrition forecasting • department headcount projections
- 04 Live dashboard • industry switcher, adaptive learning engine

200 people
SYNTHETIC TEST WORKFORCE

8 depts
ENERGY INDUSTRY SCENARIO

840
ADJUDICATED DEV SPANS

3.1%
AVG MONTHLY ATTRITION TRACKED

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Skills and tools *built* across the semester.

01

Deep *models*

CNN Image classification (L03)

RNN / LSTM

Sequence learning (L05)

Transformer

DistilBERT, ModernBERT, GLiNER

Bi-LSTM Attrition forecasting

Diffusion

DDPM image generation (L08)

DQN Cart-Pole control (L09)

02

NLP & *agents*

Named-entity recognition

10 entity types, routed ensemble

Sentence embeddings

SBERT for semantic match

Multi-agent orchestration

CrewAI + Claude

Prompting & tooling

6-scenario proprietary brain

Reinforcement learning

Policy & value methods

03

Frameworks & *tools*

PyTorch

Primary training framework

HF Transformers

Pretrained backbones

Sentence-Transformers

SBERT inference

Gymnasium RL environments

Pandas / NumPy

Data preparation

Streamlit / Plotly

Interactive dashboard

04

Practice & *craft*

Evaluation design

Adjudicated dev sidecar (840 spans)

Technical writing

Blueprints, READMEs, decks

Group collaboration

A03 & A04 group deliverables

Translating to non-experts

"DL for an 11-year-old"

Reproducible notebooks

Versioned, parameterised

• REFLECTION & NEXT STEPS

*A semester of
writing networks
by hand, training
them, and putting
two agents to work.*

— JIRI MUSIL · ITAI 2376 · SPRING 2026

➤ WHAT'S NEXT

Goals for the next semester and beyond.

- Extend Workforce Intelligence to public datasets
- Continue reinforcement learning past Cart-Pole
- ITAI 2373 (NLP) and ITAI 3377 (Edge AI and IoT)
- Open to engineering, research, and collaborator roles

GITHUB

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_Deep_Learning](https://github.com/JiriCZTX/Jiri_Musil_ITAI_2376_Deep_Learning)

CONTACT

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HCC · Applied AI & Robotics
Spring 2026